Experiment – 3

Development of an Interactive Event Page using React Hooks and State Management

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Name :

Class :

Roll No. :

Date :

Aim

Develop an using **React Hooks and State Management**, and to understand the use of useState, useEffect, and useContext for managing component state, handling side effects, and sharing data between React components.

Problem Statement

Develop an interactive Event Management Page using React.

Enhance the Event Page developed in the previous experiment by adding interactive features such as **event search, category filtering, registration status, and shared application state**.

The application should demonstrate the use of React Hooks for managing and updating data dynamically.

The following React concepts should be demonstrated:

- *useState* for managing local component state
- *useEffect* for handling side effects
- *useContext* for sharing state between components
- State management using Context API
- Event handling in React
- Conditional rendering
- Passing and updating state between components
- Dynamic rendering of event information

Or the same can be demonstrated on your project topic.

Theory Questions

1. What are React Hooks?
2. What is the purpose of useState()?
3. What is the difference between a normal variable and state in React?
4. What is useEffect()?
5. When is useEffect() executed?
6. What is the purpose of the dependency array in useEffect()?
7. What is useContext()?
8. What is Context API in React?
9. What is prop drilling?
10. How does Context API help in avoiding prop drilling?
11. What is state management?
12. What is the difference between local state and global/shared state?
13. What is conditional rendering in React?
14. How does React update the UI when state changes?
15. Why are Hooks useful in modern React applications?

Code

Add important code snippets demonstrating:

- `useState()`
- `useEffect()`
- `createContext()`
- `useContext()`
- State update functions
- Search/filter implementation
- Event registration functionality

Repository Link: _____